Student Retention Analysis

Why Students Leave, and What We Can Do About It

Prepared for the Board of Directors

by the Data Analytics Team

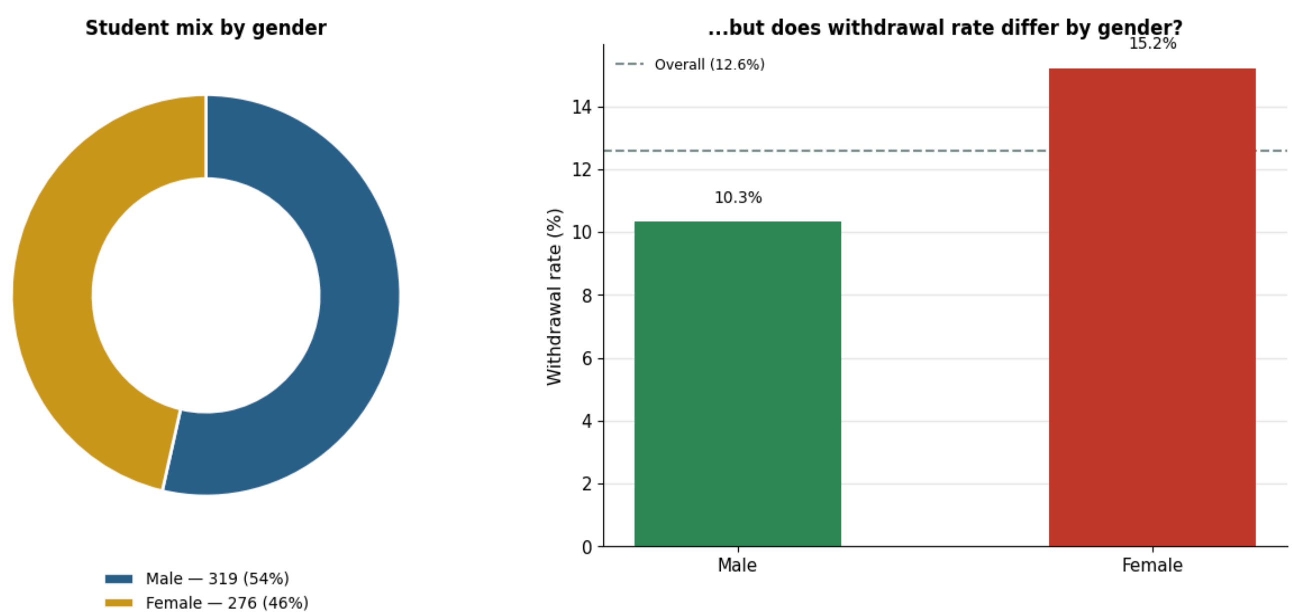
1. Purpose of This Report

This report addresses a crucial issue: why does our school have students who are leaving, and what can be done about it? To address this question, we looked at records of 580 students, attendance, payment of fees, how families enrolled in the school, and how students progressed through various grades. We did not just look at the number of students who dropped out, but instead at the early warning signs that occur prior to the student withdrawing. The aim is for the school to recognise pupils who are at risk and do something about it before it is too late. In this report, conclusions are made based on statistical analysis. Results are presented simply so that they are easily understood. Our findings are based on the evidence of our school data, not on assumptions or guesswork.

2. The Big Picture

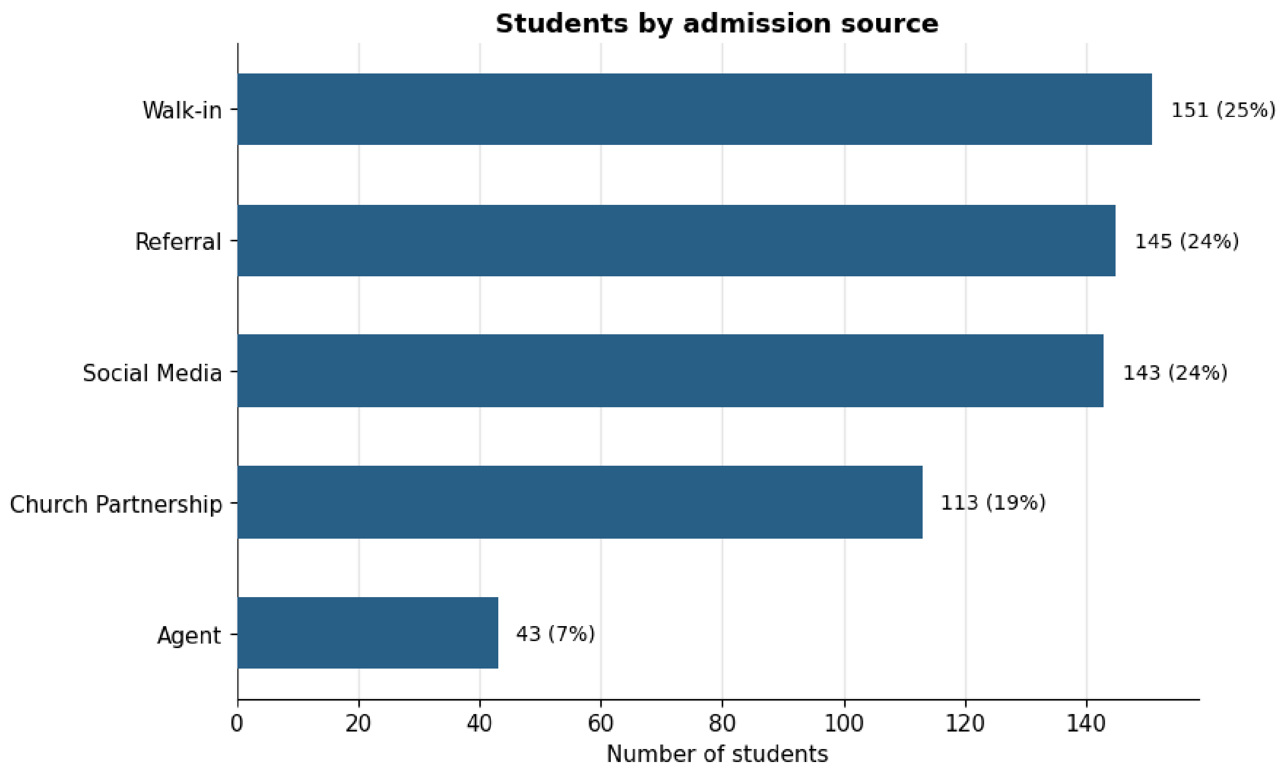
This analysis includes all 580 students. Of these, 309 (53.3%) are male, and 271 (46.7%) are female, giving a nearly even gender split.

Figure 2.0 below shows the distribution of students by gender and the withdrawal rate by gender.



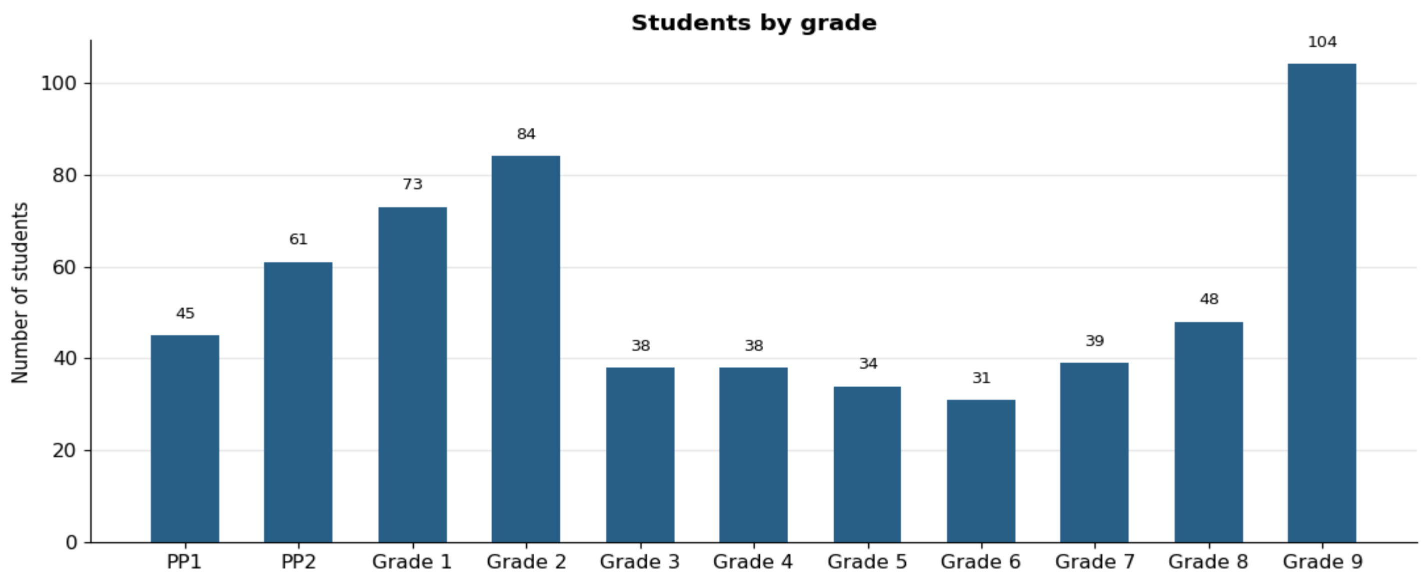
Students joined the school through five admission channels: Walk-in (148, 25.5%), Referral (143, 24.7%), Church Partnership (108, 18.6%), Social Media (138, 23.8%), and Agent (43, 7.4%).

Figure 2.1 below shows the distribution of students by admission channel and the percentage share by admission channel.



The data includes all levels of the CBC system, from PP1 up to Grade 9, where there was a higher number of learners in the lower primary grades, which are the ones that grew in this school. There have been 75 withdrawals (12.9%) from the school. That number is only a part of the picture. National statistics also reveal that dropout rates are not the same across a child's entire school experience. Dropout rates are low in the early grades of primary school but increase to approximately one in five by the end of primary, and repetition rates are also higher in the higher grades (Kenya Demographic and Health Survey ([USAID)](#)). The same pattern is found with our own data. Not all students are equally at risk for withdrawal. Some groups have withdrawal rates of less than 8%, and some are more than 70%. The remainder of this report describes those groups and their uniqueness.

Figure 2.2 below shows the distribution of students by grade and the number of students by grade.



3. Finding One: Attendance Is an Early Warning Signal

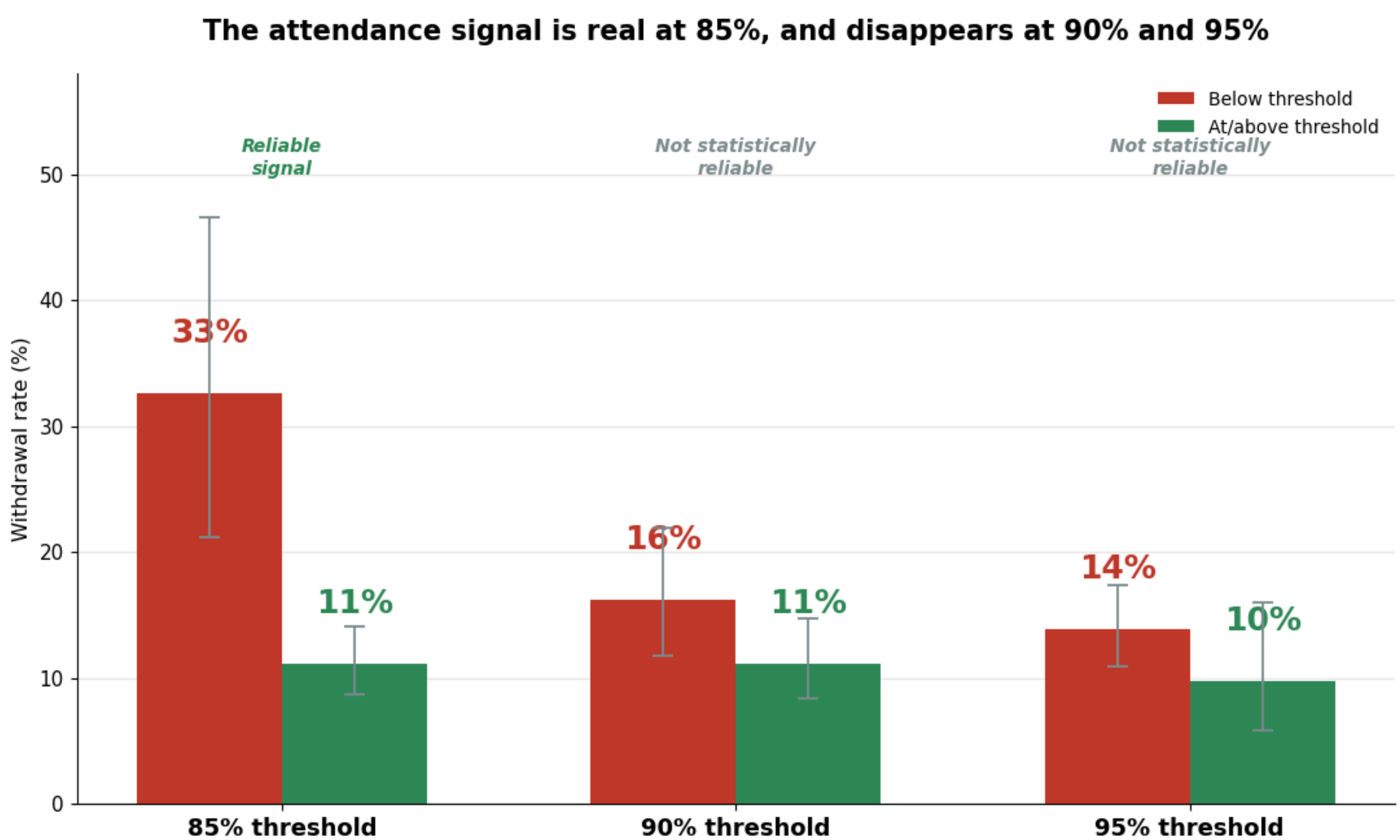
At the **85%** attendance threshold, students were **3.9 times more likely to withdraw** than students with better attendance. This result is based on **49 students**, and the analysis has **95% statistical power**, meaning we have more than enough evidence to be confident that this is a real pattern.

At the **90%** threshold, students were only **1.6 times more likely to withdraw**. Although this group included **203 students**, the statistical power dropped to **40%**, meaning the evidence is too weak to tell us with confidence that the difference is real.

At the **95%** threshold, students were only **1.4 times more likely to withdraw**. This included **447 students**, but the statistical power fell further to **25%**, giving us even less confidence that the observed difference reflects a genuine pattern rather than normal variation.

For that reason, we recommend using the 85% attendance threshold. It is the only level that provides strong, reliable evidence while remaining a simple and practical trigger for staff to monitor and act on.

Figure 3.0 below shows the distribution of students by performance threshold and the percentage share by threshold.



4. Finding Two: Fee Arrears — the Strongest Predictor We Found

We asked whether unpaid fees predict withdrawal. Using a practical cutoff of 15% of expected fees in arrears, the answer is unambiguous:

Table 4.0 below shows the distribution of students by arrears category and the withdrawal rate by arrears category.

Group	Students	Withdrawal rate
Arrears below 15%	503	8.0%
Arrears at or above 15%	77	45.5%

Of the **503** students owing **less than 15%** of their expected fees, **8.0%** withdrew. Among the **77** students owing **15% or more**, **45.5%** of them withdrew, a number almost **5.7 times higher**, making fee arrears the strongest predictor of withdrawal in this analysis. The statistical test produced a **chi-square value of 80.12** and a **p-value of 3.5×10^{-19}**. This means the chances of seeing a difference this large purely by chance are essentially zero. Students with arrears of **15% or more** were about **10 (9.56) times more likely to withdraw** than those below that level. The analysis also achieved **100% statistical power**, giving us a very high level of confidence that this is a real pattern in our data. To be clear about the two figures above: the withdrawal rate was 5.7 times higher, while the estimated odds of withdrawal were about 10 times higher. These are two different statistical measures of the same pattern, not conflicting results.

However, the risk does not suddenly appear at **15%**. It increases steadily as arrears grow. Students owing **less than 10%** of expected fees had a withdrawal rate of **7.2%**. Those owing **10% to 15%** had a rate of **9.3%**. For students owing **15% to 25%**, the rate rose to **14.6%**. Among those owing **more than 25%**, **every one of the 28 students eventually withdrew**. A family that is **12% behind** on fees, roughly a month and a half of payments, still looks much like the average family. A family that is **30% behind**, nearly a full term of unpaid fees, is at an extremely high risk of leaving. The period between these two points is where the school has the greatest opportunity to intervene.

Fee arrears are likely to reflect more than unpaid bills. They may also signal financial pressure or a family that is already beginning to reconsider staying at the school.

Figure 4.0 below shows the distribution of students by fee payment status and the withdrawal rate by fee payment status.

Implication:

Treat **15% fee arrears** as an early-warning trigger rather than simply a collections issue. As soon as a family reaches this level, staff should make supportive contact within the week to discuss payment plans or hardship options. Acting while arrears are still between **10% and 15%** is far more likely to keep a student enrolled than waiting until arrears exceed **25%**, when the outcome is often already decided.

5. Finding Three: How a Family Found Us Predicts How Long They Stay

Table 5.0 below shows the admission distribution by channels of admission.

Admission source	Students	Withdrawal rate
Referral	143	7.0%
Walk-in	148	8.1%
Church partnership	108	9.3%
Agent	43	16.3%
Social media	138	26.1%

Students who joined through **referrals** had the lowest withdrawal rate at **7.0%**. This was followed by **walk-ins (8.1%)**, **church partnerships (9.3%)**, **agents (16.3%)**, and **social media (26.1%)**. Students who joined through social media were **more than 3.7 times** as likely to withdraw as those who joined through referrals.

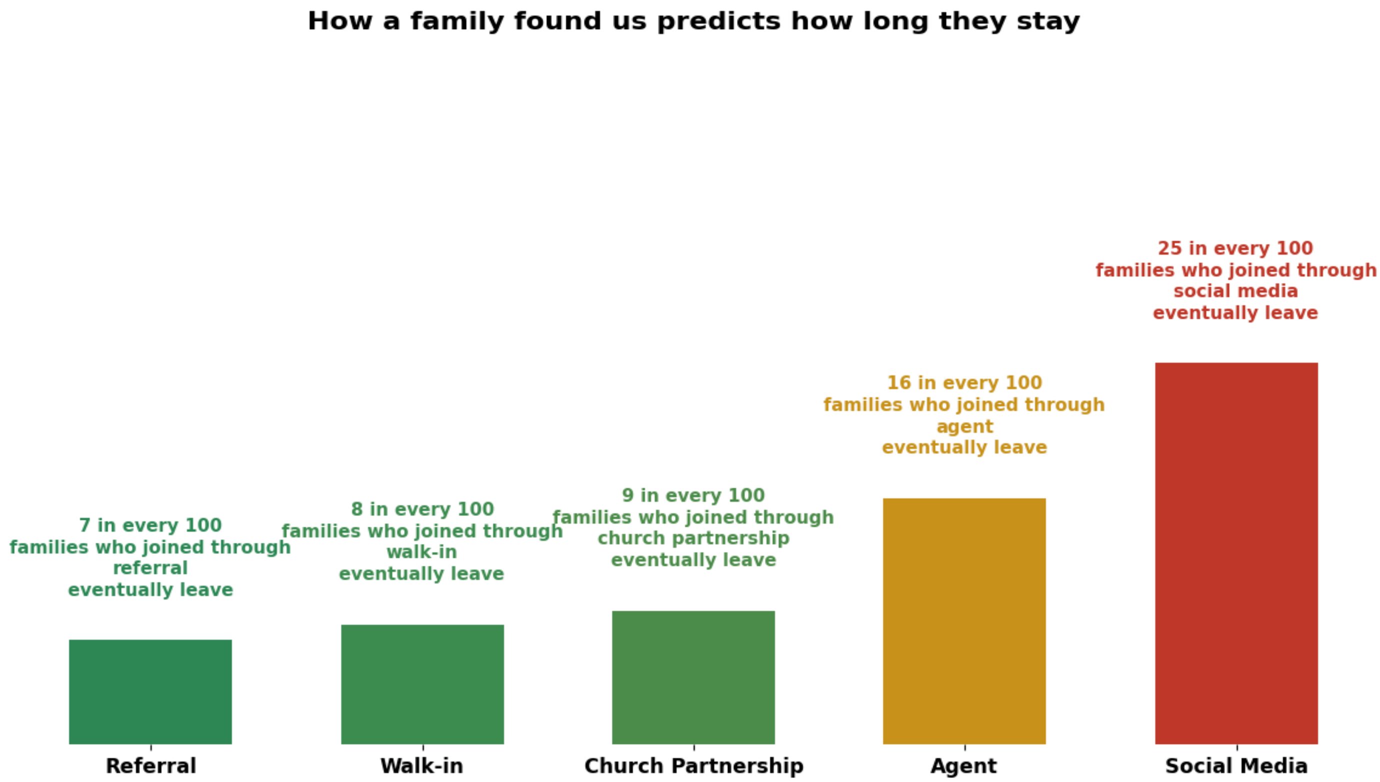
The statistical test produced a **chi-square value of 29.75** and a **p-value of 5.5×10^{-6}**, meaning there is only a tiny chance these differences happened by accident. Students who joined through social media were **4.38 times more likely to withdraw** than students who joined through referrals. This is the **"mtu wangu"** effect, and every Kenyan knows it. When your neighbour, church member, or cousin tells you, *"Hiyo shule ni poa. Mtoto wangu anasoma hapo,"* you arrive already trusting the school. For such a parent, if the bus is late one morning or the first fee statement is confusing, they are more likely to give the school the benefit of the doubt because someone they trust has already vouched for it.

Now compare that with a parent who found the school through an Instagram reel or a Facebook advert. They don't have a **"mtu wangu."** If the first term does not match what they expected, moving to another school is easy. It is like buying a phone your cousin recommends from a shop he trusts versus buying one from a Jumia advert with suspiciously good reviews. Both phones might be perfectly fine, but you are much quicker to return the second one. Agent-sourced families sit somewhere in the middle. An agent is usually rewarded for getting a child enrolled, not for making sure the family stays for years. The relationship begins as a transaction, not a personal recommendation, so it doesn't come with the same built-in trust as a referral.

Implication:

Admission source is known from the first day a family enrolls, long before attendance or fee records can identify a problem. This makes it one of the earliest opportunities to improve retention. Families joining through **social media** or **agents** should receive additional support during their first term, such as a welcome call, a structured orientation, and pairing with an existing parent family. Building trust early can help these families feel connected to the school and improve the likelihood that they remain enrolled.

Figure 5.1 below shows the distribution of students by admission channel and the withdrawal rate by admission channel.



6. Finding Four: The Grade 7/8 Transition Creates Financial Strain

Group	Students	Average arrears	Median arrears
Grade 7/8 (JSS)	86	18.9%	14.1%
Other grades	494	10.2%	8.2%

Among the **86** students in **Grade 7 and Grade 8**, the average fee arrears is **18.9%**, with a typical (median) arrears of **14.1%**. For the **494** students in all other grades, the average arrears is **10.2%**, with a median of **8.2%**. In other words, students in JSS carry **almost twice the arrears burden** of the rest of the school. Because both the average and the median are much higher, this is not being caused by just a few families with very large balances. It is a pattern seen across the JSS group.

Think about what happens when a child moves into **Grade 7**. It is not just a new classroom. Families often face the biggest fee increase in the entire school at the same time the child is adjusting to new subjects, new teachers, and, in many cases, new costs such as devices or laboratory fees. It is a bit like your landlord increasing the rent in the same month that matatu fares go up. Nothing about your daily routine has changed, but suddenly everything costs more, and the household budget feels the pressure all at once.

Finding Two already showed that **higher fee arrears are strongly linked to students leaving the school**. This means the Grade 7 transition is more than just a more expensive year. It is one of the points where financial pressure and the risk of withdrawal come together.

Implication:

This gives the school a clear opportunity to act **before** problems begin. Reach out to every Grade 6 family before their child moves to Grade 7. Explain the new fees in **actual shillings**, not just percentages, and discuss payment options before the first JSS invoice is issued. A conversation in **November** about **January's** fees cost the school very little. Trying to recover a family that is already **25% behind** a few months later is much harder and, as Finding Two showed, is often too late.

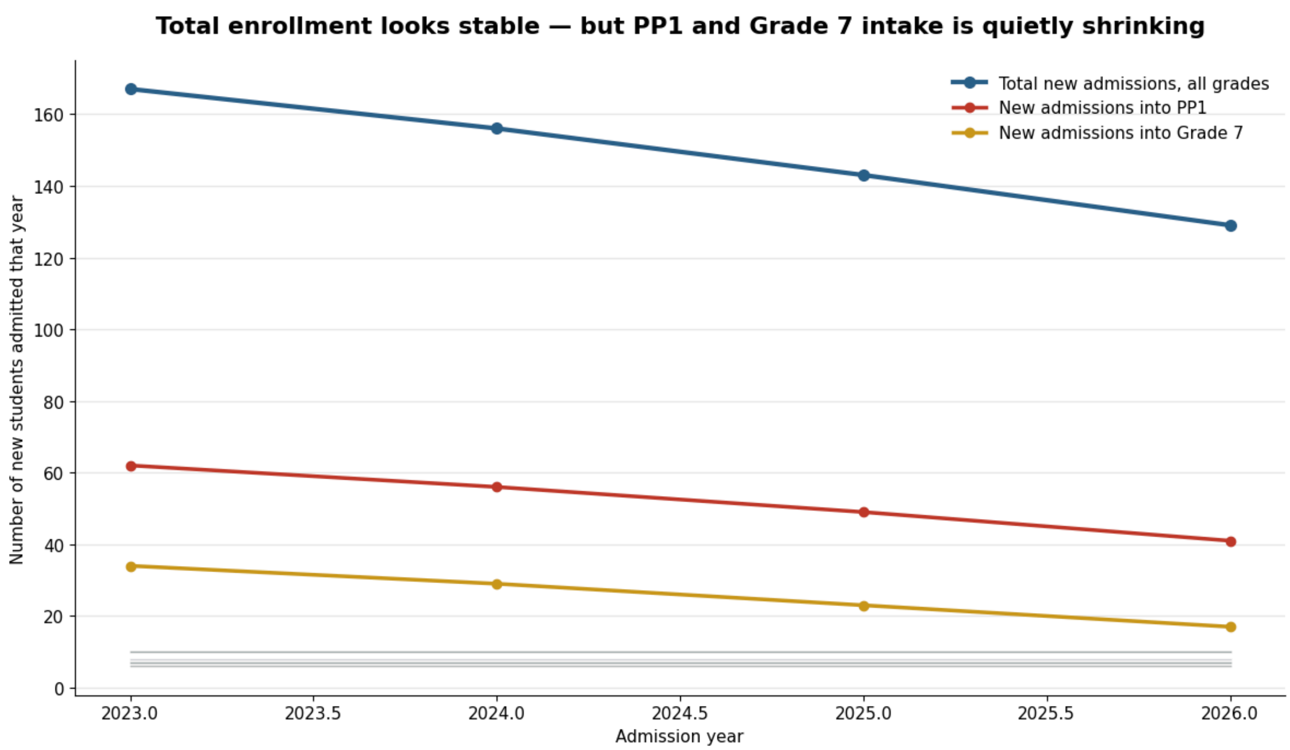
7. Finding Five: New Admissions Are Not Shrinking Evenly

The table below summarizes the year-over-year change in new admissions by grade.

Grade	Change in admissions/year	Statistically significant?
PP1	`−7.0 students/year`	Yes ($p = 0.002$)
Grade 7	`−5.7 students/year`	Yes ($p = 0.0009$)
All other grades	No meaningful change	No

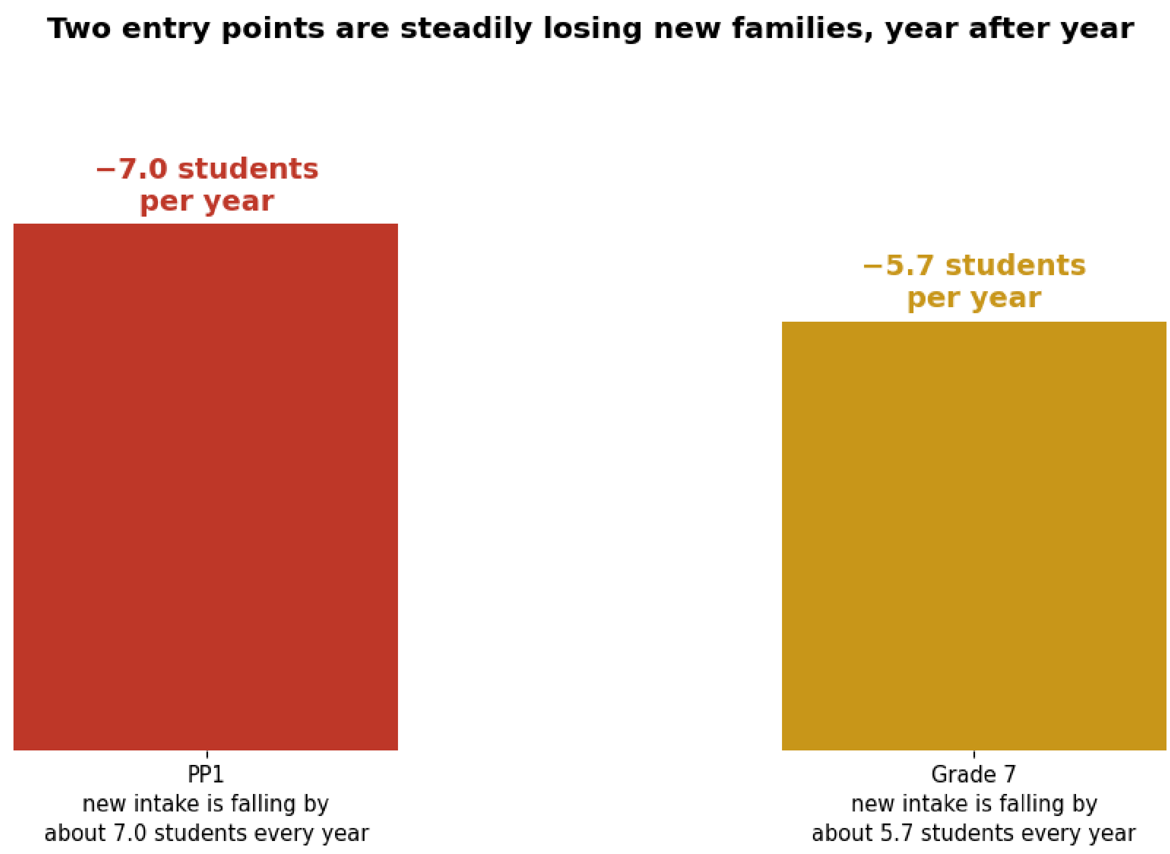
New admissions into **PP1** are falling by about **7 students every year**, while **Grade 7** is losing about **6 students every year**. The statistical tests gave **p-values of 0.002** for PP1 and **0.0009** for Grade 7. This means there is **very strong evidence that these declines are real, not just normal year-to-year fluctuations**. The **R^2 values of 0.996 and 0.998** also show that these downward trends have been remarkably consistent over time. No other grade shows a meaningful increase or decrease.

Figure 7.0 below shows the distribution of new admissions by admission year and the number of new admissions into PP1 and Grade 7 by admission year.



At first glance, the school's total enrollment still looks healthy. But that headline number hides what is happening underneath. It is like watching the water flowing out of a river without noticing that it has stopped raining upstream. The river still flows for a while, but sooner or later the shortage catches up. In the same way, fewer children joining at **PP1** and **Grade 7** may not affect total enrollment immediately, but a few years later, those smaller groups move through the school, and every grade behind them becomes smaller too.

Figure 7.1 below shows the distribution of new admissions by entry point and the annual decline in new admissions by entry point.



It is also worth noticing that these are not just any two grades. They are the school's **main entry points**. **PP1** is where many families first join the school. **Grade 7** is where Junior Secondary begins, and as Finding Four showed, it is also the point where fee arrears increase sharply. These are the two moments when families are most likely to compare schools and ask themselves whether to join or whether to stay.

Implication:

Track new admissions by **grade**, not just the total number of students. The overall enrollment figure can stay stable for several years while the entry points quietly shrink. If **PP1** and **Grade 7** continue to decline, the school should focus its recruitment efforts on those two grades and ask why families are choosing other schools at exactly those points. The reasons could include fees, competition, location, or simply word of mouth not reaching as many families as it once did.

8. Do These Findings Hold Up Together, or Are They Overlapping?

So far, we have looked at each factor on its own. The next question is whether these factors are **independent**, or whether some are simply pointing to the same underlying problem.

When we analyzed **all the factors together**, **fee arrears** were the **only factor that remained a strong predictor of withdrawal**. The model gave an **adjusted odds ratio of 1.13 for every 1% increase in fee arrears**, with a **p-value below 0.001**. This means there is **very strong evidence** that higher fee arrears are linked to a higher chance of a student leaving, even after taking attendance, grade level, admission source, and length of time at the school into account.

Attendance, JSS status, admission source, and student tenure all looked important when examined on their own. However, once fee arrears were included in the same analysis, **they no longer added meaningful information**. Their effects largely overlapped with what the fee arrears were already telling us.

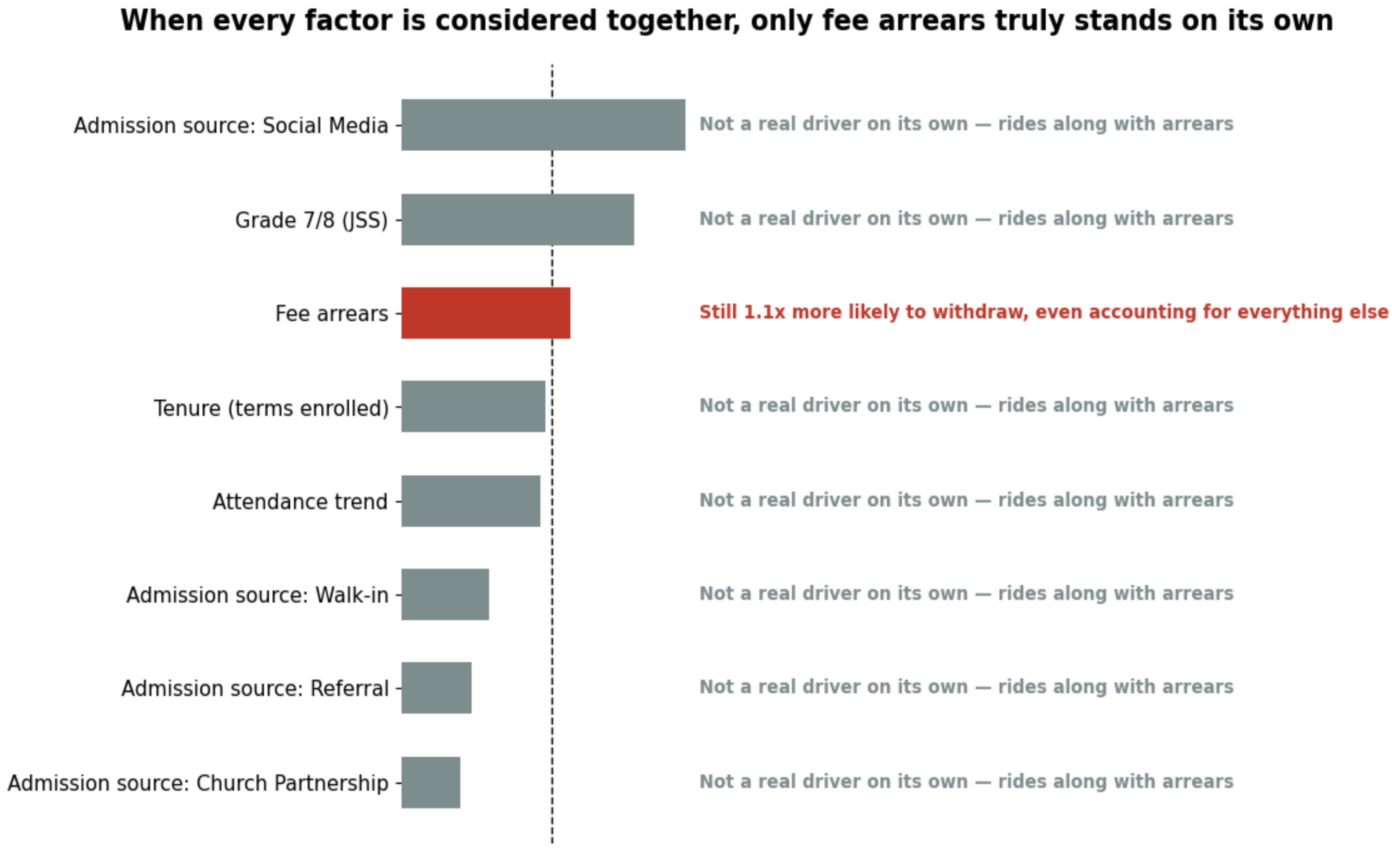
Think of a gym membership. Someone who has not been to the gym in weeks is more likely to cancel. Someone paying for an expensive premium membership is also more likely to cancel. If you look at each factor by itself, both seem to predict cancellation. But when you look at them together, you may find that the expensive membership is the real driver, while poor attendance is simply one of the first signs that the person is already thinking about leaving.

The same pattern appears in our data. Families under financial pressure may begin missing payments first. As that pressure grows, attendance may start to slip, and eventually the student withdraws. That is why attendance and JSS status seem to predict withdrawal on their own but add little once fee arrears are taken into account.

Implication:

Financial pressure appears to be the main factor behind several of the patterns in this report. Supporting families with fee arrears is therefore likely to improve more than just payment rates—it may also help improve attendance and overall student retention, rather than requiring separate interventions for each issue.

Figure 8.0 below shows the factors affecting student withdrawal and whether each factor is an independent driver of withdrawal.



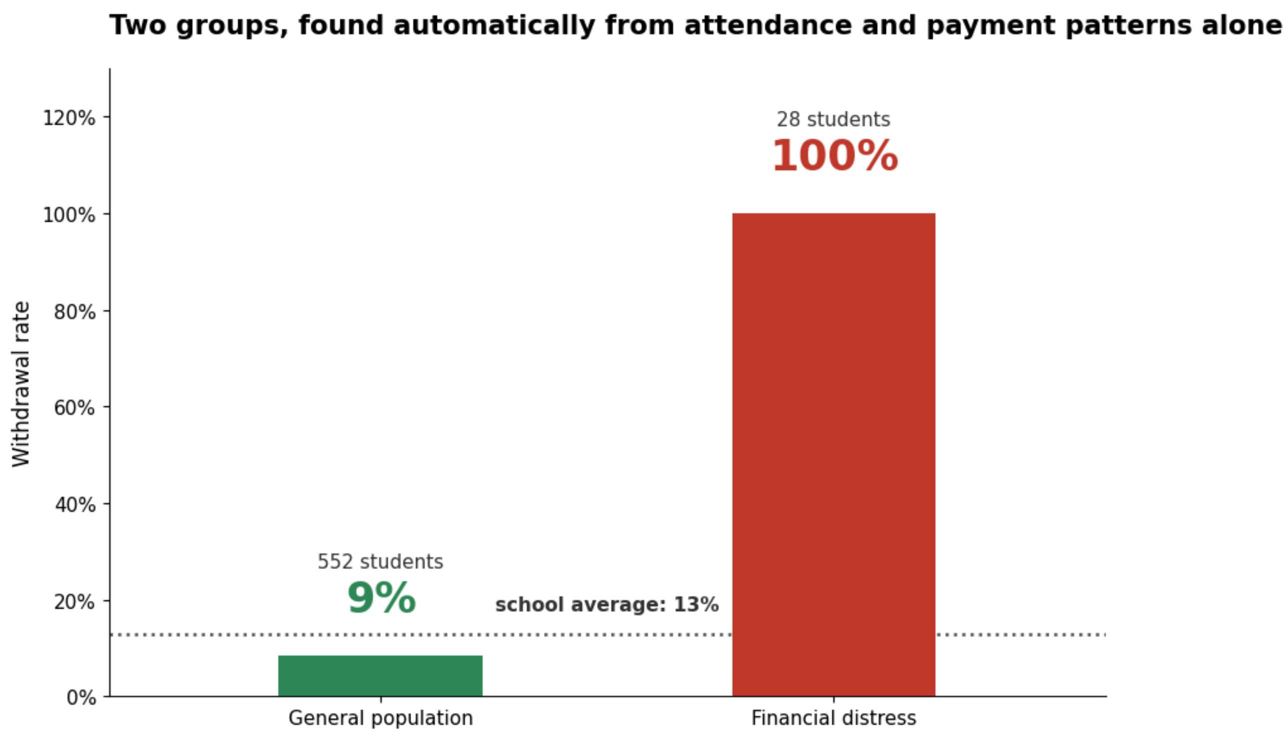
9. Two Very Different Kinds of At-Risk Students

Instead of deciding in advance what an "at-risk" student should look like, we let the data group students based on their own patterns. The clearest result was **two groups**.

The first group contains **552 students**, with a withdrawal rate of **9%**. The second is a much smaller group of **28 students**, and **every one of them eventually withdrew**.

At first, you might expect the smaller group to have much poorer attendance. Surprisingly, that isn't the case. Their average attendance is almost the same as everyone else's, **92% compared with 91%**. The real difference is **fee arrears**. Students in the larger group owed an average of **9%** of their expected fees, while the smaller group owed an average of **62%**.

Figure 9.0 below shows the distribution of students by risk group and the withdrawal rate by risk group.



This is important because it confirms what we found earlier. The computer was **never told which students withdrew**. It simply grouped students with similar patterns together. Yet it still separated the same high-risk group identified in Finding Eight. In other words, the data reached the same conclusion in two different ways: **fee arrears are the strongest warning sign**.

Students who joined through **Social Media** and **Agents** appeared more often in this high-risk group than expected, which is consistent with Finding Three.

Implication:

The evidence points to one clear priority. Rather than creating separate intervention programmes for different types of at-risk students, the school should focus first on identifying and supporting families with growing fee arrears. The early-warning system recommended in Finding Two would capture almost all of the students in this highest-risk group.

10. Recommendations and Next Steps

No.	Action	Owner	Timing
1	Automated 15% arrears alert + proactive family contact	Finance / IT	This term
2	Separate protocols for attendance risk vs. financial risk	Academic / Student Support	Next term
3	Proactive JSS-transition financial conversations	Finance / Admissions	Before the next Grade 7 cohort
4	Enhanced onboarding for social-media / agent families	Admissions	Next admissions cycle
5	Grade-level intake monitoring dashboard	Data / IT	Ongoing

We recommend a formal review after one year to assess the impact of these five actions and adjust the approach based on what the next round of data shows.

11. Conclusion

Student withdrawal is not random. It follows clear patterns that can be identified early using the school's own data. This analysis shows that **fee arrears are the strongest and most reliable early warning sign**. While attendance, admission source, and the Grade 7 transition all help identify students who may be at risk, much of their effect is explained by the financial pressure families are already experiencing. The good news is that this gives the school a clear opportunity to act before students leave. A supportive conversation when fee arrears begin to build, particularly around the **15%** mark and before the **Grade 7 transition**, is likely to have the greatest impact on retention. Combined with stronger onboarding for families joining through social media and agents and regular monitoring of admissions into PP1 and Grade 7, these actions provide a practical, evidence-based strategy for improving student retention and strengthening the long-term growth of the school.

For further detail or additional analysis, please contact the Data Analytics Team.